

🚀 What is SEO and Why It Matters in Digital Marketing

Understanding Digital Marketing

Digital marketing refers to all the strategies and techniques businesses use to promote products, services, or content online. This includes channels like social media, email, paid advertising, websites, videos, and more. The primary goal is to reach potential customers where they spend most of their time: online.

SEO, or **Search Engine Optimization**, is a key component of digital marketing. Its purpose is to make your website visible on search engines such as Google, Bing, or Safari. When someone searches for a product, service, or answer to a question, SEO helps your website appear in those search results, ideally on the first page. For example, if a user searches for “best laptops under \$500,” effective SEO can position your site so that they see your page first and are likely to click through.

Why SEO Matters

SEO is crucial for several reasons:

1. **Most users don’t go past page one:** Websites on the first page of search results receive the majority of clicks.
2. **It generates organic traffic:** Unlike paid ads, SEO brings visitors to your website for free.
3. **It reveals customer intent:** SEO helps you understand what potential customers are searching for, allowing you to meet their needs.
4. **It builds credibility and long-term growth:** High-ranking sites are often seen as more trustworthy, and consistent SEO improves visibility over time.

SEO is not a single task but a set of practices designed to help search engines **understand**, **trust**, and **rank** your content effectively.

□ How Search Engines Work

Search engines operate in three main steps:

1. **Crawling:** Search engines send bots to visit every page on your website. Think of this like a librarian reading every book in a library.
2. **Indexing:** Once crawled, search engines store and organize your content in a database, similar to how a librarian catalogs books.
3. **Ranking:** Finally, search engines decide which pages best answer the user’s query and display them in order of relevance and quality.

The better your SEO practices, the higher your pages rank when people search for relevant terms.

📖 Section 1 — Keyword Research (Understanding What People Search For)

What Are Keywords?

Keywords are the words and phrases people type into search engines when looking for information, products, or services. Examples include:

- “What is SEO”
- “Best SEO tools”
- “SEO tutorial for beginners”
- “Cheap running shoes Nairobi”

Keywords act as clues to what users want, helping you tailor your content to their needs.

Why Keyword Research Matters

Conducting keyword research allows you to:

- Understand exactly what people are searching for.
- Avoid guessing what users want.
- Create content that aligns with real search demand.
- Identify opportunities where you can compete effectively.

Keyword Research Process

Step 1: Start with Seed Keywords

Seed keywords are broad, simple words or phrases that define your topic. They act as your starting point. Examples:

- SEO basics
- SEO tools
- Keyword research

Step 2: Expand Keyword Ideas

Use tools to discover more specific search terms.

Free & Beginner-Friendly Tools:

- **Google Autocomplete:** Type a keyword and see suggestions. Example: typing “SEO” may suggest “SEO for beginners.”
- **Related Searches & People Also Ask:** Look at suggestions at the bottom of Google results and common questions.
- **Forums & Q&A Sites:** Reddit and Quora often show real user questions.
- **Google Keyword Planner:** Provides search volume (requires a Google Ads account).
- **Google Trends:** Shows popularity of terms over time.
- **Ubersuggest / Keyword Surfer:** Simple tools for keyword ideas.

Paid Tools for More Advanced Research:

- Ahrefs
- SEMrush
- Moz Keyword Explorer

Step 3: Measure Keyword Metrics

Each keyword should be evaluated for:

Metric	Meaning	Why It Matters
Search Volume	How many people search for it monthly	Indicates potential traffic
Keyword Difficulty	How hard it is to rank for	Helps prioritize achievable keywords
Search Intent	The reason behind the search	Ensures your content matches what the user wants

Step 4: Understand Search Intent

Search intent is the user’s purpose behind a search query:

- **Informational:** Looking for knowledge (e.g., “what is SEO”).
- **Commercial:** Comparing options (e.g., “best SEO tools”).
- **Transactional:** Ready to act or purchase (e.g., “buy SEO course online”).

Aligning your content with search intent improves engagement and conversions.

Step 5: Prioritize Your Keywords

Not all keywords are equal. Choose based on: relevance, volume, difficulty, intent, and business goals. Organize your findings in a spreadsheet with columns for keyword, volume, difficulty, intent, priority, and notes.

Keyword Research Assignment

1. Pick a topic (e.g., “healthy snacks blog”).
 2. Find 50 keyword ideas using free or paid tools.
 3. Record metrics in a spreadsheet.
 4. Write a brief analysis explaining:
 - Why you selected your seed keywords
 - Which tools you used
 - How you determined search intent
 - Your top 5 target keywords
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Section 2 — On-Page SEO (Optimizing Each Page)

What Is On-Page SEO?

On-page SEO involves optimizing elements **on a webpage** so that search engines can understand the content, and users can find value. Effective on-page SEO improves:

- Visibility in search results
- Click-through rates
- Overall rankings

Key On-Page Elements

1. Title Tag

The title of your page as seen in search results. It should:

- Include the main keyword
- Be catchy
- Stay under ~60 characters

2. Meta Description

The snippet under the title in search results. It should:

- Describe the page
- Include the keyword
- Encourage clicks

3. Heading Tags (H1, H2, H3)

These structure content like chapters in a book:

- H1: Main page title
- H2/H3: Subheadings

4. URL Structure

The web address of your page should be:

- Short
- Descriptive
- Include hyphens (e.g., `website.com/beginners-seo-guide`)

5. Internal Linking

Linking to other pages within your site helps both users and search engines explore content. Example: “Learn more about SEO tools here.”

6. Content Optimization

Content should:

- Answer the user’s question
- Be written clearly
- Use keywords naturally (avoid keyword stuffing)

7. Image Optimization

- Compress images for faster loading
- Use descriptive file names
- Include alt text for accessibility and SEO

8. Mobile-Friendly Design

Most users search on mobile devices. Ensure your website looks good and loads fast on phones.

On-Page SEO Checklist

Task	Done
Title optimized	☐
Meta description	☐
H1 with keyword	☐
Clean URL	☐
Good content	☐
Internal links	☐
Image alt text	☐
Mobile friendly	☐

On-Page SEO Assignment

1. Choose a page (blog post or product).
 2. Apply the on-page SEO steps.
 3. Complete the checklist.
 4. Write a short report summarizing your changes.
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Section 3 — Technical SEO (Site Structure & Speed)

What Is Technical SEO?

Technical SEO ensures that your website is **crawlable, indexable, fast, secure, and user-friendly**. It lays the foundation for other SEO efforts and directly impacts rankings and user experience.

Key Technical SEO Concepts

1. Crawl & Index

- **Crawl:** Search engines scan your pages to understand your content.
 - **Index:** Pages are stored in the search engine's library.
Practical Test:
 - Use **Google Search Console (Free)** → **Coverage Report** to see which pages are indexed.
 - Check **robots.txt** in DevTools → Network tab to ensure pages aren't blocked.
 - Interpretation:
 - ☐ Pages in index → good.
 - ☐ Missing important pages → fix sitemap or crawling rules.
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2. XML Sitemap

- A file listing all important pages to help search engines discover content.
Practical Test:
 - Access <https://yourdomain.com/sitemap.xml>.
 - Use **Screaming Frog SEO Spider Free** to crawl your site and check sitemap URLs.
 - Interpretation:
 - ☐ All main pages listed.
 - ☐ Missing pages → add them to the sitemap.
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3. robots.txt

- Tells search engines which pages **not** to crawl (e.g., admin pages).
Practical Test:
 - Visit <https://yourdomain.com/robots.txt>.
 - Use DevTools → Network → Refresh to see if the file loads.
 - Interpretation:
 - ☐ Correctly blocks private pages.
 - ☐ Blocking important pages → adjust rules.
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4. Site Speed & Performance

Site speed is crucial for **user experience, Core Web Vitals, and rankings**.

Key Metrics & How to Analyze (Free Tools):

Metric	Description	How to Test	Interpretation
Largest Contentful Paint (LCP)	When the main content loads	Chrome DevTools → Performance tab → Record → Look for LCP marker. Also: PageSpeed Insights	☐ ≤2.5s → good, ☐ 2.5–4s → needs improvement, ☐ >4s → poor
First Contentful Paint (FCP)	When the first text/image appears	Same as LCP	Faster FCP → better perceived speed
Cumulative Layout Shift (CLS)	Visual stability	PageSpeed Insights or DevTools → Performance → Experience	☐ <0.1 → good, ☐ 0.1–0.25 → needs improvement, ☐ >0.25 → poor
Time to First Byte (TTFB)	Server response time	DevTools → Network → Look at document request → TTFB	☐ <200ms → optimal, ☐ 200–500ms → moderate, ☐ >500ms → slow server
Long Tasks / JS Blocking	JS slowing page	DevTools → Performance → Look for long tasks (yellow bars)	Reduce heavy JS scripts
Resource Load Sizes	Large files slow pages	DevTools → Network → Size column	Compress images, minify CSS/JS

Practical Tests for SEO Speed:

1. Open Chrome → F12 → **Performance Tab** → Record → Reload page.
2. Check **LCP, FCP, CLS, Long Tasks**.
3. Use **PageSpeed Insights** (<https://pagespeed.web.dev/>) for mobile/desktop audit.
4. Use **GTmetrix Free** for waterfall chart analysis.

5. HTTPS Security

- Secure websites are trusted by users and search engines.
Practical Test:
- Look for **https://** and padlock in browser.
- DevTools → Security tab → Check certificate validity.
- Interpretation:
 - ☐ HTTPS active → secure, ☐ mixed content → fix, ☐ HTTP → migrate.

6. Structured Data (Schema)

- Extra code to help search engines understand your content.
Practical Test:

- Use **Google's Rich Results Test (Free)**: <https://search.google.com/test/rich-results>
 - Enter a page URL → Validate structured data.
 - Interpretation:
 - ☐ Valid structured data → rich results possible.
 - ☐ Errors → fix JSON-LD or Microdata.
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7. Canonical Tags

- Avoid duplicate content by indicating the main page version.
Practical Test:
 - DevTools → Elements → Search for `<link rel="canonical" href="...">`.
 - Interpretation:
 - ☐ Correct canonical → good.
 - ☐ Missing/incorrect → duplicate content risk.
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8. Mobile Friendliness

- Google ranks mobile-optimized sites higher.
Practical Test:
 - DevTools → Toggle device toolbar (Ctrl+Shift+M) → Check layout & usability.
 - Google Mobile-Friendly Test (<https://search.google.com/test/mobile-friendly>).
 - Interpretation:
 - ☐ Layout adapts → good.
 - ☐ Buttons too small → UX issue.
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Technical SEO Checklist with Practical Tests

Task	How to Test	Done
Site crawlable	Google Search Console Coverage	☐
XML sitemap exists	<code>/sitemap.xml</code> + Screaming Frog	☐
robots.txt correct	<code>/robots.txt</code> + DevTools Network	☐
Site loads fast	Chrome DevTools, PageSpeed Insights	☐
Core Web Vitals (LCP/FCP/CLS)	DevTools → Performance → Record	☐
Mobile friendly	DevTools → Device Toggle + Google Mobile-Friendly Test	☐
HTTPS active	Browser padlock + DevTools Security tab	☐

Task	How to Test	Done
Structured data	Google Rich Results Test	☐
Canonical tags correct	DevTools → Elements → <code><link rel="canonical"></code>	☐

□ Pro Tips

- Always combine **lab tests** (DevTools, PageSpeed Insights) with **field data** (Google Search Console, Chrome UX Report).
 - Focus on **visible content** first (LCP) — fastest way to improve perceived speed.
 - Fix **blocking JS and large images** first; these usually give the biggest LCP improvement.
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Tools Every SEO Beginner Should Know

1. Browser Developer Tools (F12)

- **Purpose:** Inspect HTML, CSS, and JavaScript of your website. Check how elements are rendered and debug issues.
- **Key Features:**
 - **Mobile Responsiveness:** Simulate different screen sizes to ensure your site is mobile-friendly.
 - **Page Speed Analysis:** Identify slow-loading elements like large images or scripts.
 - **Console Errors:** Detect JavaScript errors that may affect UX or SEO.
- **Tip:** Chrome DevTools also allows you to audit accessibility and performance, helping you optimize for both users and search engines.

2. Google Search Console (GSC)

- **Purpose:** Monitor your site's presence in Google search results.
- **Key Features:**
 - Track which keywords bring traffic and which pages are indexed.
 - Identify crawl errors or security issues.
 - Submit sitemaps and monitor backlinks.
- **Tip:** Use the "Performance" report to see CTR, impressions, and position trends for your keywords.

3. Google Analytics

- **Purpose:** Understand visitor behavior, traffic sources, and conversions.
- **Key Features:**
 - Analyze which pages get the most engagement.
 - Identify traffic sources (organic, paid, referral, social).
 - Track conversions to measure the success of SEO and marketing campaigns.
- **Tip:** Integrate Google Analytics with GSC for a more complete SEO insight.

4. PageSpeed Insights

- **Purpose:** Evaluate your website's speed and performance.
- **Key Features:**
 - Provides scores for mobile and desktop performance.
 - Offers actionable recommendations (e.g., compress images, remove render-blocking scripts).

- **Tip:** Page speed affects both SEO rankings and user experience—fast pages keep visitors engaged.
 - 5. **Screaming Frog SEO Spider**
 - **Purpose:** Desktop tool that crawls your entire site like a search engine bot.
 - **Key Features:**
 - Finds broken links, duplicate content, and missing metadata.
 - Helps generate XML sitemaps.
 - Analyze page titles, headers, and response codes.
 - **Tip:** Essential for auditing large sites where manual checks aren't practical.
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💡 SEO and Online Advertising Connection

SEO is not just about organic search; it strengthens your paid campaigns. Here's how:

1. **Choosing the Right Keywords**
 - Ads perform better when keywords are aligned with user intent.
 - Use SEO keyword research tools like **Ahrefs** or **SEMRush** to find high-converting search terms.
2. **Optimizing Meta Tags and Landing Pages**
 - Search engines reward pages with clear titles, meta descriptions, and structured content.
 - Well-optimized landing pages improve ad relevance and reduce bounce rates.
3. **Enhancing Google Ads Quality Score**
 - Ads linked to well-optimized pages often cost less per click.
 - Quality Score depends on expected CTR, ad relevance, and landing page experience.
4. **Content Relevance**
 - Matching ad copy with SEO content ensures consistent messaging.
 - Users are more likely to engage and convert when the landing page fulfills the promise of the ad.
5. **Data-Driven Improvements**
 - Insights from GSC and Google Analytics can refine both SEO and ad campaigns.
 - Example: If a page ranks high organically but has low conversions, update content or CTA to improve performance.

In short, integrating SEO with paid campaigns helps businesses **maximize visibility, lower costs, and improve ROI**.

📖 Keyword Research Tools for Beginners and Pros

1. Google Autocomplete (Free)

- **What it does:** Shows popular search suggestions when you type a word or phrase in Google.
- **Why it's useful:** Reveals what people are actively searching for.
- **How to use it:**
 1. Go to [Google.com](https://www.google.com).
 2. Start typing a topic (e.g., "healthy snacks").
 3. See the suggestions that appear in the dropdown.

4. Note these down as potential keywords.
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2. Google Related Searches (Free)

- **What it does:** Shows related queries at the bottom of Google search results.
 - **Why it's useful:** Gives additional keyword ideas you might not think of.
 - **How to use it:**
 1. Search for a keyword on Google.
 2. Scroll to the bottom of the results page.
 3. Write down the related searches.
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3. Google “People Also Ask” Box (Free)

- **What it does:** Shows frequently asked questions related to your keyword.
 - **Why it's useful:** Gives ideas for content topics and long-tail keywords.
 - **How to use it:**
 1. Search your seed keyword in Google.
 2. Look for the “People Also Ask” box.
 3. Expand questions and note them down.
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4. Google Keyword Planner (Free with Google Ads Account)

- **What it does:** Shows keyword search volume, competition, and trends.
 - **Why it's useful:** Lets you choose keywords based on data, not guesswork.
 - **How to use it:**
 1. Sign in to [Google Ads](#).
 2. Go to **Tools** → **Keyword Planner** → **Discover new keywords**.
 3. Enter a seed keyword (e.g., “SEO basics”).
 4. Review keyword suggestions with metrics:
 - Avg. monthly searches
 - Competition (low/medium/high)
 - Top of page bid (optional)
 5. Export your list to a spreadsheet.
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5. Google Trends (Free)

- **What it does:** Shows how popular a keyword is over time and compares search interest across regions.
- **Why it's useful:** Identify trending keywords and seasonal patterns.
- **How to use it:**
 1. Go to [Google Trends](#).
 2. Enter a keyword (e.g., “healthy snacks”).

3. Check interest over time and by country.
 4. Compare multiple keywords to see which is more popular.
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6. Google Search Console (Free)

- **What it does:** Shows which keywords your website is already ranking for.
 - **Why it's useful:** Helps optimize existing pages and find low-hanging keyword opportunities.
 - **How to use it:**
 1. Go to [Google Search Console](#).
 2. Select your website.
 3. Click **Performance** → **Queries**.
 4. See the keywords, impressions, clicks, and CTR.
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7. Ubersuggest (Free/Paid)

- **What it does:** Suggests long-tail keywords, search volume, and difficulty scores.
 - **How to use it:**
 1. Go to [Ubersuggest](#).
 2. Enter a keyword.
 3. Review suggested keywords and metrics.
 4. Export to spreadsheet for analysis.
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8. Keyword Surfer (Free Chrome Extension)

- **What it does:** Shows keyword search volume and related keywords directly in Google search results.
 - **How to use it:**
 1. Install the [Keyword Surfer extension](#).
 2. Search a keyword on Google.
 3. See volume and related keywords instantly in the sidebar.
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9. Paid Advanced Tools (Optional for Later Use)

These are industry-standard tools for deep research:

- **Ahrefs:** Keyword Explorer, competition analysis, and content gap research.
 - **SEMrush:** Keyword Magic Tool, search intent, and CPC data.
 - **Moz Keyword Explorer:** Search volume, difficulty, and SERP analysis.
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Practical Student Exercise Using Google Tools

1. **Pick a topic:** Example: “home workout equipment.”
2. **Generate seed keywords** using Google Autocomplete.
3. **Expand keywords** with Related Searches + People Also Ask.
4. **Check search volume & competition** in Google Keyword Planner.
5. **Analyze trends** using Google Trends to see which keywords are growing.
6. **Document findings** in a spreadsheet:

| Keyword | Source Tool | Monthly Searches | Competition | Trend | Notes |

7. **Prioritize Top 5 Keywords** for content creation based on relevance, search volume, and intent.

Technical SEO & Core Web Vitals Terms Explained

SEO (Search Engine Optimization) – The practice of improving a website so that it ranks higher in search engine results pages (SERPs). Effective SEO increases organic traffic, improves visibility, and builds trust with users.

Digital Marketing – Strategies used to promote products, services, or content online. Includes SEO, social media, email marketing, paid ads, video content, and more. SEO is one key component of digital marketing.

Organic Traffic – Visitors who come to your website naturally through search results, not paid ads. SEO aims to maximize this traffic.

Keywords – Words or phrases users type into search engines to find information. Keywords reveal user intent and help shape content strategy. Examples: “SEO tools,” “best laptops under \$500.”

Seed Keywords – Broad, foundational keywords that define a topic. Used as a starting point for keyword research. Example: “SEO basics.”

Long-Tail Keywords – Longer, more specific search queries, often with lower search volume but higher intent. Example: “cheap running shoes Nairobi.”

Search Intent – The purpose behind a user’s search query. Can be **informational** (seeking knowledge), **commercial** (comparing options), or **transactional** (ready to buy or act). Aligning content with search intent improves engagement and conversions.

Search Volume – The number of times a keyword is searched for in a month. Indicates potential traffic opportunity.

Keyword Difficulty / Competition – A measure of how hard it is to rank for a keyword. High difficulty means more competition; low difficulty offers easier ranking opportunities.

Meta Description – The snippet below a page title in search results. Summarizes the page content and can include a keyword to improve click-through rate (CTR).

Title Tag – The clickable page title in search results. Should include main keywords and be compelling to attract clicks.

Heading Tags (H1, H2, H3) – HTML elements that structure content. H1 is the main title, H2/H3 are subheadings. Proper structure helps search engines understand the content.

URL Structure – The web address of a page. Should be short, descriptive, and use hyphens instead of spaces. Example: website.com/beginners-seo-guide.

Internal Linking – Links from one page on your website to another. Helps users navigate and helps search engines understand site structure.

Content Optimization – Writing content that answers user questions clearly, uses keywords naturally, and provides value without keyword stuffing.

Image Optimization – Reducing image file size for faster loading, using descriptive filenames, and adding alt text for SEO and accessibility.

Mobile-Friendly / Responsive Design – Ensuring websites display correctly on mobile devices, with readable text, tappable buttons, and fast loading.

Google Autocomplete – Google search feature suggesting popular queries as you type. Useful for discovering keyword ideas.

Google Related Searches – Suggestions at the bottom of search results showing similar queries. Helps expand keyword research.

People Also Ask (PAA) – Google search box showing commonly asked questions related to a keyword. Valuable for identifying long-tail keywords and content ideas.

Google Keyword Planner – Free tool (with Google Ads account) that provides search volume, competition, and keyword trends to guide content strategy.

Google Trends – Shows keyword popularity over time and by region. Useful for identifying seasonal trends or growing topics.

Google Search Console (GSC) – Free tool from Google to monitor site performance in search results. Shows indexed pages, keyword rankings, clicks, impressions, CTR, and coverage errors.

Ubersuggest – Keyword tool that suggests related keywords, search volume, and difficulty scores.

Keyword Surfer – Chrome extension showing keyword search volume and related keywords directly in Google search results.

Ahrefs / SEMrush / Moz Keyword Explorer – Paid tools for advanced keyword research, competition analysis, content gap analysis, and tracking rankings.

CTR (Click-Through Rate) – The percentage of users who click a link in search results after seeing it. High CTR indicates compelling titles and meta descriptions.

Landing Page Optimization – Improving a page's content, layout, and SEO elements so it converts visitors effectively, especially for paid ads or targeted traffic.

Quality Score (Google Ads) – Metric that evaluates ad relevance, landing page experience, and expected CTR. Better SEO on the landing page can improve Quality Score and reduce ad costs.

SERP (Search Engine Results Page) – The page displayed by a search engine after a user performs a search. SEO aims to get your pages to appear as high as possible on the SERP.

Content Relevance – How closely a page’s content matches a user’s query or intent. High relevance improves rankings and user engagement.

Conversion – When a user completes a desired action, like signing up, buying, or downloading. SEO can help drive traffic that converts.

Data-Driven SEO – Using analytics (Google Analytics, GSC) to understand user behavior, traffic sources, and search performance to optimize pages.

Practical Free Tools for SEO Testing – Chrome DevTools (F12) for technical checks, PageSpeed Insights for speed, GSC for indexing and performance, Google Mobile-Friendly Test for mobile optimization, and Rich Results Test for structured data.

Crawl – The process by which search engines visit your web pages to understand their content. If a page isn’t crawlable, search engines won’t see it and it won’t appear in search results.

Index – After crawling, search engines store your page in their library. Only indexed pages can appear in search results.

XML Sitemap – A file listing all your important pages to help search engines find and crawl them. Think of it as a roadmap of your site.

robots.txt – A file telling search engines which pages or sections of your site to avoid crawling. Useful for private pages, admin areas, or duplicate content.

Site Speed – How fast your site loads. Faster sites improve user experience and SEO. It’s affected by images, scripts, CSS, caching, and server response time.

HTTPS Security – Secure websites using HTTPS are trusted by users and search engines. HTTPS prevents data from being intercepted and is a ranking factor.

Structured Data / Schema – Extra code (like JSON-LD or Microdata) added to your pages to tell search engines what the content means, for example marking a product so Google can show price and ratings in search results.

Canonical Tags – Tags that tell search engines the “main” version of a page, preventing duplicate content issues.

Mobile Friendliness – Pages optimized for mobile devices. Includes responsive design, readable text, and buttons that are easy to tap.

Largest Contentful Paint (LCP) – A performance metric measuring the time it takes for the largest visible content on the page (like a big image, video, or text block) to load. Good LCP is ≤ 2.5 seconds.

First Contentful Paint (FCP) – Measures when the first text or image appears on the screen, giving users a visual signal that the page is loading.

Cumulative Layout Shift (CLS) – Measures how much the layout moves unexpectedly while the page loads. Low CLS (<0.1) means stable content, high CLS frustrates users.

First Input Delay (FID) – The time from when a user first interacts with the page (like clicking a button) to when the browser responds. It measures responsiveness.

Time to First Byte (TTFB) – The time the server takes to start sending data after a user requests a page. Faster TTFB improves both user experience and SEO.

PerformanceObserver / PerformanceEntry – JavaScript APIs used to record LCP, FCP, and other performance metrics programmatically.

Resource Timing – Tracks download times for scripts, images, CSS, and other resources. Helps identify slow-loading assets.

Navigation Timing – Measures key stages of page load, including request start, response start, and load event.

Element Timing – Lets you measure render times of specific elements, useful if the largest element isn't the most important.

Lazy Loading – A technique where images or videos load only when they come into the viewport. Improves initial page speed.

Compression (Gzip / Brotli) – Reduces file sizes of resources sent from the server to the browser.

Minifying CSS/JS – Removes unnecessary characters from code (like spaces and comments) to reduce file size and improve load speed.

HTTP Status Codes – Signals from the server: 200 (OK), 301/302 (redirects), 404 (not found), 500 (server error).

Web Fonts / Font Loading – Custom fonts can delay text rendering, affecting FCP and LCP.

Viewport / Visibility / CSS Properties – Elements outside the viewport, with `opacity: 0`, or hidden using `display: none` are ignored by LCP calculations.

Long Tasks / JS Blocking – JavaScript tasks that take a long time to execute and block rendering. Slows LCP and FCP.

Background Images / Hero Sections – Large visible images or sections that significantly affect LCP.

Back/Forward Cache – Browser feature that stores pages when navigating back or forward. Can affect LCP measurements if not considered.

Cross-Origin / Timing-Allow-Origin – Security restrictions on measuring render times of images or resources from other domains. Setting Timing-Allow-Origin allows accurate performance tracking.